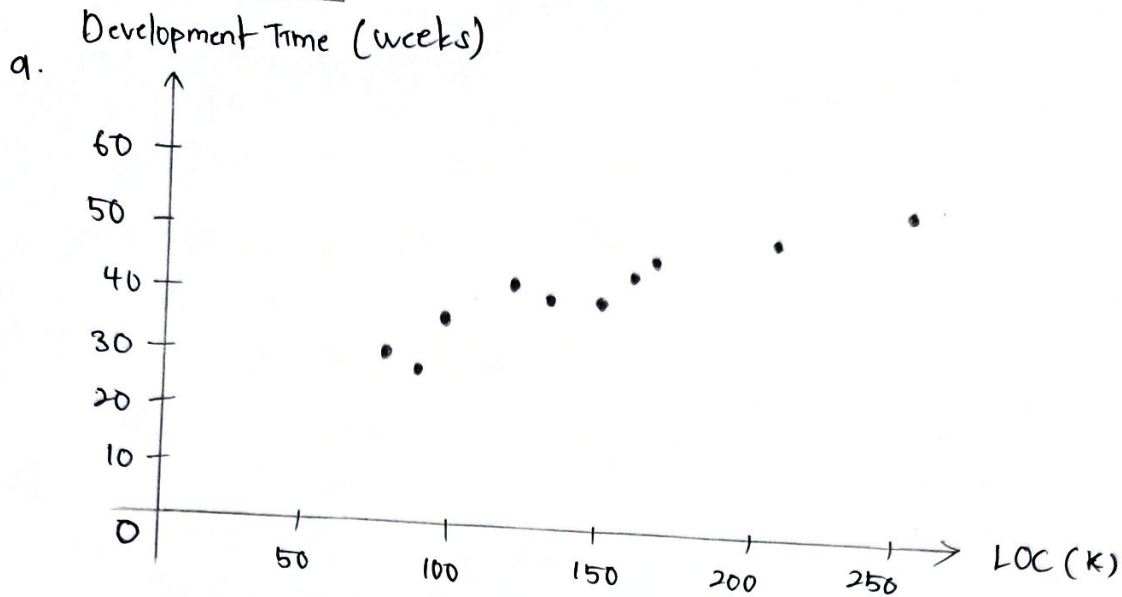


Question 5



b.

	LOC x	Development times y	xy	x^2	y^2
1	150	40	6000	22500	1600
2	100	35	3500	10000	1225
3	200	50	10000	40000	2500
4	80	30	2400	6400	900
5	170	45	7650	28900	2025
6	120	38	4560	14400	1444
7	160	42	6720	25600	1764
8	90	28	2520	8100	784
9	250	55	13750	62500	3025
10	130	37	4810	16900	1369
$\Sigma x =$	1450	$\Sigma y =$ 400	$\Sigma xy =$ 61910	$\Sigma x^2 =$ 235300	$\Sigma y^2 =$ 16636



$$r = \frac{\sum xy - \left(\frac{\sum x \sum y}{n} \right)}{\sqrt{\left[\left(\sum x^2 \right) - \frac{(\sum x)^2}{n} \right] \left[\left(\sum y^2 \right) - \frac{(\sum y)^2}{n} \right]}}$$

$$= \frac{61910 - \frac{(1450)(400)}{10}}{\sqrt{\left[(235300) - \frac{(1450)^2}{10} \right] \left[(16636) - \frac{(400)^2}{10} \right]}}$$

$$= \frac{3910}{\sqrt{(25650)(636)}}$$

$$= 0.98$$

$$r = \frac{3910}{\sqrt{(25650)(636)}} = 0.98$$

c. $\hat{y}$ = development time

x = LOC

$$\hat{y} = b_0 + b_1 x$$

$$b_1 = \frac{\sum xy - \left(\frac{\sum x \sum y}{n} \right)}{\sum x^2 - \frac{(\sum x)^2}{n}}$$

$$= \frac{61910 - 58000}{235300 - 210250}$$

$$= 0.156$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

$$= \frac{400}{10} - 0.156 \left(\frac{1450}{10} \right)$$

$$= 17.38$$

$$\therefore \hat{y} = 17.38 + 0.156 x$$

$$d. \quad R^2 = \frac{SSR}{SST}$$

$$SSR = \sum (\hat{y} - \bar{y})^2$$

$$= (40.78 - 40)^2 + (32.98 - 40)^2 + (48.58 - 40)^2 + (29.86 - 40)^2 + (43.9 - 40)^2 + (36.1 - 40)^2 + (42.34 - 40)^2 + (31.42 - 40)^2 + (56.38 - 40)^2 + (37.66 - 40)^2$$

$$= 610.3$$

$$SST = (40 - 40)^2 + (35 - 40)^2 + (50 - 40)^2 + (30 - 40)^2 + (45 - 40)^2 + (38 - 40)^2 + (42 - 40)^2 + (28 - 40)^2 + (55 - 40)^2 + (37 - 40)^2$$

$$= 636$$

$$R^2 = \frac{610.3}{636}$$

$$= 0.96$$

$$e. \quad \hat{y} = 17.38 + 0.156(180) \\ = 45.46 \text{ weeks}$$

$\therefore$ The prediction is 45.46 weeks